OPERATION

Temperature Controller System

Product Description

The digital microprocessor temperature controller is designed to provide temperature control of refrigerators or freezers. The controller also provides a constant readout of the sample temperature inside of the unit. A touch keypad allows the user to easily select the display units, set point, and differential set point.

⚠ **Please Note: The digital temperature controller has been factory set and tested to allow your unit to operate at its desired temperature cycle.**

Adjusting the settings on the controller will alter these factory settings. WE STRONGLY RECOMMEND YOU CONTACT THE MANUFACTURER'S TECHNICAL SUPPORT DEPARTMENT BEFORE MAKING ANY ADJUSTMENTS TO THIS CONTROLLER. TECH SUPPORT PHONE NUMBER IS (800) 648 4041 Option 4

CHECK TEMPERATURE HISTORY

Press and release [UP] button. The display will show the maximum temperature ever reached since the last reset.

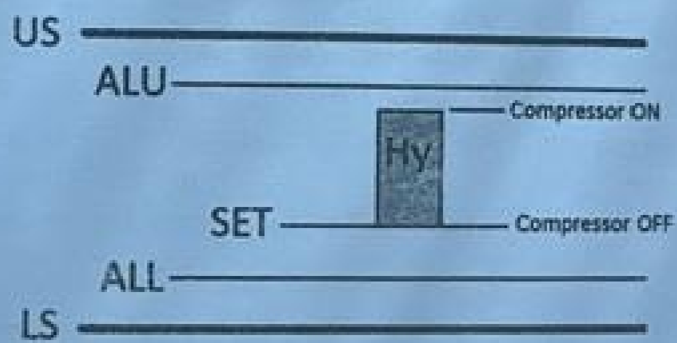
Press and release [DOWN] button. The display will show the minimum temperature ever reached since the last reset.

Press and hold [SET] for more than 3 seconds, while the maximum or minimum temp is displayed. (rSt message will be displayed).

CHECK THE SET POINT

Press and release [SET] button. The display will show the current set point value.

Operation



During the normal operation, the refrigerator's (or freezer's) compressor would turn on and off, in order to maintain the cold temperature in the storage chamber.

In this controller, the point where the compressor is cut off is called "SET POINT". The point where the compressor is turned on is calculated by adding the value of "SET POINT" and "Hy" (temp differential).

For example, if you wish to maintain the operation temperature between 3°C and 7°C, you would set "SET" = 3°C, and "Hy" = 4°C.

"ALU" is the high temp alarm point, and "ALL" is the low temp alarm point. Both alarm settings will alert users when the refrigerator's (or freezer's) temp is out of range, via visual & audible alarm, and remote alarm contact.

"US" is the upper setting limit, and "LS" is the lower setting limit. Both limit settings will prevent users accidentally adjust "SET", "ALU", or "ALL" outside the range.

CODE	DESCRIPTION	FACTORY SETTING
SET	Temp set point (compressor off point)	3°C or 38°F (refrigerator) -24°C or -12°F (freezer)
Hy	Temp differential between compressor start and off point	3°C or 5°F (NOT RECOMMENDED TO CHANGE)
ALL	Low temp alarm point	1°C or 34°F (refrigerator) -28°C or -18°F (freezer)
ALU	High temp alarm point	10°C or 50°F (refrigerator) -15°C or 5°F (freezer)
Lod	Screen display choice (air or sample probe)	P3
CF	Celsius & Fahrenheit unit change	
O3	Sample (display) probe calibration / offset	0
OT	Air (control) probe calibration / offset	0
US	The maximum limit that SET or ALU could reach	10°C or 50°F (refrigerator) -10°C or 14°F (freezer)
LS	The minimum limit that SET or ALL could reach	1°C or 34°F (refrigerator) -30°C or -22°F (freezer)

<u>Change the set point (compressor turn-off point)</u>
Press and hold [SET] until °C or °F icon blinking. Press [UP] or [DOWN] to change the setting value. Then, press [SET] once to confirm the new setting.

<u>Change the other settings:</u>
Press and hold both [SET] and [DOWN] at the same time until "Hy" appears on the display.

Press [UP] or [DOWN] to scroll different settings. Press [SET] to enter the setting. Press [UP] or [DOWN] to change value. Press [SET] once to confirm the new setting. The display will show the next setting.

At any setting, press and hold both [SET] and [UP] to exit out the setting mode, or simply leave the display alone for 10 seconds.

Change the readout from °C to °F, or °F to °C

Press and hold the [LIGHT] icon button for 5 seconds. The controller will restart and change the display scale from °C to °F, or °F to °C.

Advanced Settings – for service technician only

⚠ ATTENTION: This section is for service technicians or experienced users only. Altering the following settings can result in malfunction or inaccurate temperature readout.

Air and Sample Temperature Display

The controller has the capability to display either the air or sample temperature readout. For the normal operation, the sample-simulated temperature (P3) is displayed in order to provide users the content temperature. For the actual operation, the air temperature (P1) is used to control the compressor's cycle.

This is a useful tool for you to make a precise adjustment, or temperature validation process.

"Lod" setting allows you to display either air (P1) or sample (P3). Press and hold both [SET] and [DOWN] at the same time until "Hy" appears on the display.

Press [UP] or [DOWN] until "Lod" shows up. Press [SET] to enter the setting. Press [UP] or [DOWN] to toggle between the air temp "P1", or the sample temp "P3". Press [SET] once to confirm the new setting. The display will now show the temp you have selected.

⚠ We strongly recommend you to change the "Lod" setting back to "P3" before you complete the service. This will allow users to see the sample-simulated temperature, and the controller will be able to alert when the sample temp is out of range.

Calibration / offset

"OT" setting allows you to change the air probe's calibration. "O3" setting allows you to change the sample probe's calibration.

Please be sure you have a NIST traceable and calibrated thermometer. Place your thermometer's probe next to our sensor accordingly, air vs. air, or sample vs. sample, before making an adjustment on either "OT" or "O3".

For more advanced settings, please contact our Technical Service Department for assistance. (800) 648 4041 Option 4

Quick Troubleshooting Guide

Check these items before calling for service

PROBLEM:	POSSIBLE CAUSE / SOULTIONS:
Unit does not run	• Electrical circuit is not 110-120V 60Hz. • The power cord is not plugged in. • No power at electrical outlet. Check to make sure breaker is not tripped or fuse is not blown. Additionally, make sure unit is not plugged into a Ground Fault Circuit Interrupter (GFCI) type of outlet.
Unit does not maintain at the proper temperature	• Check the room temperature. We recommend the refrigerator or freezer should be placed in an air conditioned room between 65°F to 85°F. If the room temp is too warm, the refrigerator or freezer may not be able to maintain the interior temp at proper range. • Door is not closed properly. • Amount of stored product is overloaded. • Product replacements are pushed against rear wall or interrupted the proper refrigerator air circulation. For the proper air circulation, place the products evenly on each shelf. Do not push against the refrigerator's rear or side walls. • Evaporator is blocked by frost or ice. Remove the products, unplug the refrigerator or freezer power, and allow the unit to defrost. If the problem still exists, call for service. • 3rd party thermometer is placed incorrectly. For proper temperature monitoring, the thermometer should be place in the middle of refrigerator. **PLEASE NOTE!** Prior to shipment, each refrigerator and freezer has been calibrated and tested at proper temperature range.

Appliance runs too long	• Prolong door openings. • Control set too cold. • Room temperature is high which will make the unit work harder to keep cool.
Temperature of external wall surface is warm	• The exterior walls can be as much as 30 degrees warmer than room temperature due to the embedded condenser coils. This is normal when the unit is operating.
Compressor noises	• Compressor may be overheated. Please check the room temp and ensure the range is within 65°F to 85°F. If the problem still exists, call for service.
Moisture collects inside	• Door gasket is not sealing properly. Check for debris, cracks, and items passing through door at the gasket. • The refrigerator or freezer is facing a doorway or is underneath of air conditioning vent. Relocate the unit or redirect air vent. • Too many door openings. Minimize time door is open. • Hot, humid weather increases condensation. • Make sure there is a water trap (U-shaped loop) in the drain tube near the compressor. This will "trap" a small amount of water in the loop and prevent air from entering the chamber through the tube.
Moisture collects on outside surface	• Hot, humid weather increases condensation. • As humidity decreases, moisture will disappear.
Odor inside the unit	• Interior needs to be cleaned. See section on maintenance and cleaning in this manual. • Make sure product containers are tightly sealed to prevent leakage
Door will not close	• The unit is not level. Refer to the Leveling section at the beginning of this manual • Check for dirt and debris or items passing through the door seal.

MOISTURE DURING THE SUMMER SEASON

The amount of moisture, condensation, or high humidity related issues increase during the summer and, in most cases, will self-resolve when the weather cools down. Please note a refrigeration system will NOT generate moisture or water but simply condenses the moisture that is already in the chamber. Keeping the unit in an air conditioned, low humidity space will resolve many issues. Other things you should check

1. Location of the refrigerator (See Quick Troubleshooting Guide above)
2. Door sealing and frequency of door opening event (See Quick Troubleshooting Guide above)
3. Make sure there is a water trap (U-shaped loop) in the drain tube near the end This will "trap" a small amount of water in the loop and prevent air from enter the chamber through the tube.

BEFORE CALLING THE MANUFACTURER'S TECHNICAL SUPPORT DEPARTMENT, please have the unit's model and serial number ready as well as the problem description. The model and serial number is located on the serial tag which can be found on the interior left upper wall of the unit.

CLEANING

PART	CLEANING AGENTS	TIPS AND PRECAUTIONS
Interior and Door Liners	Soap and water Baking soda and water	Use 2 tablespoons of baking soda in 1 quart of warm water Be sure to wring excess water out of sponge or cloth before cleaning around controls, light bulb or any electrical parts.
Door Gaskets	Soap and water	Wipe gaskets and their seating surfaces with a clean soft cloth
Shelves	Soap and water	Do not wash removable shelves in dishwasher
Exterior and Handles	Soap and water Non Abrasive Glass Cleaner	Do not use commercial household cleaners, ammonia, or alcohol to clean handles Use a soft cloth to clean smooth handles Do not use a dry cloth to clean smooth handles

Clean the glass with a mild detergent and water on a soft cloth or sponge. Rinse with water and wipe dry.

For Swinging door units, pay particular attention to the gasket and its seating surfaces. Any debris buildup on these can cause air leaks into the compartment resulting in condensation as well as reduced efficiency.

Hydrocarbon Service Notes

According to U.S. Code of Federal Regulation 40 Part 82, this refrigerator employs the natural refrigerant, specifically hydrocarbon, R290 or R600a.

Because of the nature of hydrocarbon refrigerant, for mechanical repair, such as recharge the refrigerant, or compressor replacement, should only be carried out by a certified refrigeration technician.

The safety of this equipment is listed by Underwriter Laboratory (UL) under Standard 471, Section SB – "natural refrigerant".